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Long-Term Outcomes After Proton Beam Radiation Therapy for Intraocular Vascular Tumors

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Purpose/Objective(s): To present long-term follow-up for patients with intraocular vascular tumors treated with proton beam radiation therapy (PBRT).

Materials/Methods: We performed a review of our prospectively maintained database for all patients with intraocular vascular tumors treated with PBRT at our institution between 1994 and 2012. We report patient demographics, tumor characteristics, dose schemes, treatment responses, and complications. Patients with at least 6 months of follow-up data were included in the study.

Results: Our review identified 29 evaluable patients (29 eyes) with vascular tumors of the eye, with the most common diagnosis being choroidal hemangioma (n = 22). Additionally, three diffuse choroidal hemangiomas associated with Sturge-Weber syndrome, three retinal angiomas, and one hemangioblastoma associated with von Hippel-Lindau syndrome were also identified. The main analysis hereafter focuses on the choroidal hemangioma patients (n = 22), with mean follow-up time of 78.5 months (range 12-177 months, median 69.2 months). Median age at diagnosis was 48.3 years (range 17-72). All 22 patients were symptomatic at the time of diagnosis. Initial tumor thickness, as measured by ultrasonography, ranged from 2.0 mm to 7.2 mm (mean 3.8 mm). All patients except one had the anterior tumor border located posterior to the equator. Patients were treated using total doses between 18 and 24 Cobalt Gray Equivalent (CGE) in four fractions. Best corrected visual acuity improved or remained stable in 20 of 22 eyes (91%). Of the 16 eyes initially presenting with retinal detachment, 15 experienced reattachment (94%). Cystoid macular edema was initially documented in 16 eyes and completely resolved in 13 eyes (81%). Subretinal fluid was initially present in 17 eyes and resolved in 16 eyes (94%). Tumor thickness decreased in all eyes as of last examination. Notable complications occurred in five patients, with the most common being radiation retinopathy and atrophy; anti-VEGF therapy was used for treatment. Nine patients received full dose to the disc and macula (range 18-24 CGE), none of whom had significant worsening of vision. The additional seven intraocular vascular tumors with variable histologies have also demonstrated tumor response after PBRT at mean follow-up of 29.5 months (median 24.4 months) with stable or improved visual acuity.

Conclusions: This series demonstrates consistent and long-term tumor response and visual preservation in patients with benign vascular tumors of the eye, including choroidal hemangiomas, diffuse hemangiomas with Sturge-Weber syndrome, retinal angiomas, and hemangioblastoma related to von Hippel-Lindau syndrome. The data presented represents the longest available follow-up in the literature to date for choroidal hemangiomas treated with PBRT.

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Ruthenium Eye Plaques for Uveal Melanoma: Long-Term Clinical Outcomes

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Purpose/Objective(s): We have previously reported a dosimetric advantage for Ruthenium 106 (Ru¹⁰⁶) versus COMS Iodine 125 (I¹²⁵) in select patients with small-medium size uveal melanoma (UM). Here we examine the long-term disease control rates and toxicities for patients treated with Ru¹⁰⁶ plaques.

Materials/Methods: We identified 47 patients with primary UM treated with Ru¹⁰⁶ eye plaque brachytherapy from 2005-2009. Bebig "Plaque Simulator" software was used for both Ru¹⁰⁶ and I¹²⁵ treatment-planning. Dose distributions were calculated for a prescription of 85 Gray (Gy) to the tumor apex. Patients were selected for Ru¹⁰⁶ based on reduced dose to vision critical structures (optic disc and macula) and followed clinically per institutional protocol (every 3 months for year 1 postoperatively, every 6 months for years 2-5, and annually thereafter) with ophthalmologic examination/ultrasonography (US) of the eye. Biannual US of the liver and liver panel were performed as part of metastatic surveillance. Patients were monitored for local control (LC), disease free survival (DFS), and overall survival (OS), as well as for radiation retinopathy and optic neuropathy (ON). Kaplan-Meier analyses were performed for LC, DFS, OS, and toxicity rates. Univariate analyses with log-rank test and Cox multivariate analyses were performed to identify factors predicting for DFS and visual toxicities.

Results: Median ophthalmologic follow-up was 58.9 months. Median age of patients was 64.3 years (range [R]: 26.3-91.1 years). Median apical height of tumors was 3 mm (R: 1.5-5.2 mm) and median basal size was 10 mm (R: 4.5-17.0 mm). Globe preservation was achieved in 96% of patients. 2 patients underwent enucleation and 1 patient had repeat brachytherapy for local recurrence. 5-year LC was 89% (95% CI = 78.7%-99.1%) with no failures reported after 24.9 months. 5-year DFS (freedom from local or distant failure) was 78% (95% CI = 65.2%-90.6%). 15 patients developed radiation retinopathy, with a 5 year rate of 34%; and, 4 patients developed radiation ON, with a 5 year rate of 14.3%. Basal tumor size > 10 mm was an independent predictor for poor DFS (HR = 1.679; 95% CI = 1.041-2.708; p = 0.034). Distance (≤11.61 mm vs > 11.61 mm; p = 0.008) and dose (≤5.81 Gy vs > 5.82 Gy; p = 0.001) to the optic disc and distance (≤11.76 mm vs > 11.76 mm; p = 0.001) and dose (≤7.93 Gy vs > 7.94 Gy; p = 0.007) to the macula predicted the development of radiation retinopathy or ON.

Conclusions: In the largest reported United States' series for UM patients treated with Ru¹⁰⁶, excellent long-term LC and DFS rates were achieved. Toxicities were less frequent than those reported in I¹²⁵ series, emphasizing that the use of Ru¹⁰⁶ in a carefully selected patient population provides great long-term clinical outcomes with less toxicity. Ru¹⁰⁶ has a viable role in the treatment of UM, supporting its re-introduction in U.S. practice.

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Radiation Tolerance of the Optic Pathway After Fractionated Proton Radiation Therapy

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Purpose/Objective(s): Radiation-induced optic neuropathy (RION) is a concern when fractionated radiation doses to the optic pathway exceed 54-60 Gy. Tolerance is thought to be comparable between photon and proton radiation therapy. We examined the rate of RION following proton radiation therapy to the cranial base.

Materials/Methods: This retrospective study included patients with chordoma or chondrosarcoma of the base of skull treated with proton therapy between 1985 and 2013, who received a minimum of 30 Gy(RBE) to any part of the anterior optic pathway defined as the chiasm or either optic nerve.

Results: Thus far, we identified 103 patients who met inclusion criteria. Median age was 44 years. Sixty of 103 (58.3%) had chordoma. Median total dose was 72.0 Gy (RBE) (range: 59.4-83.6) delivered in a median of 37 fractions (range: 26-63). Median target volume was 16.0 cc (range: 1.9-109.0). Median follow-up time was 84 months. Median maximum radiation dose to the optic chiasm was 61.9 Gy (RBE) (range: 10-69.0), and median radiation dose was 57.4 (1.3-62.3). Median maximum radiation dose to the optic nerves was 58.9 (10.0-77.3), and median radiation dose was 22.6 (1.5-65.4). Following radiation therapy, eight of 103 patients (7.8%) had improved vision. Twenty-seven patients (26.2%) reported